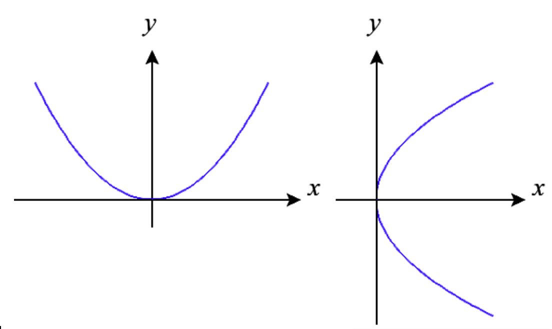
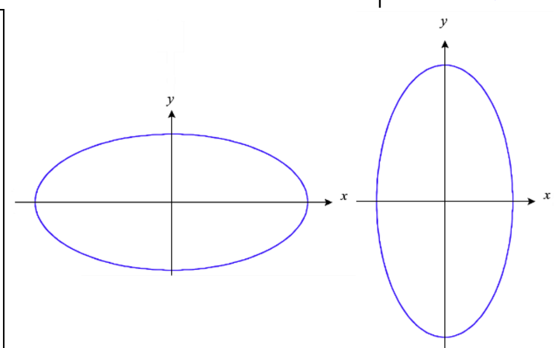
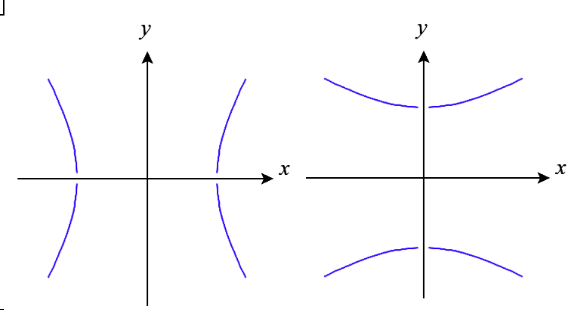
MTH 202 Chapter 1: Three-Dimensional Space & Vectors

1.7 Quadric Surfaces

Throughout Chapter 1 we have been trying to build up intuitions about 3-space and how to create some basic types of curves (lines) and surfaces (Cylindrical surfaces & planes). In this section we will attempt to use our knowledge of conic sections (a topic that is discussed in various previous courses, most commonly Algebra II, Precalculus, or Calculus II) to help understand how to visualize/recognize several new surfaces that we call Quadric surfaces.

<u>Recap of Conic Sections</u>

Name of Conic Section	**Equation(s)**	**Graphs**
Parabola	$y=a(x-h)^2+k$ *or* $x=a(y-k)^2+h$	
Ellipses	$\dfrac{(x-h)^2}{a^2}+\dfrac{(y-k)^2}{b^2}=1$	
Hyperbolas	$\dfrac{(x-h)^2}{a^2}-\dfrac{(y-k)^2}{b^2}=1$ *or* $\dfrac{(y-k)^2}{a^2}-\dfrac{(x-h)^2}{b^2}=1$	

Quadric surfaces can be thought of as the generalization of conics in three dimensions. In general, the equation of a quadric surface looks like: $Ax^2+By^2+Cz^2+Dxy+Eyz+Fxz+Gx+Hy+Iz+J=0$, but we will only work with very simple forms of this equation in MTH 202. In our class you will be expected to be able to identify quadric surfaces by name and sketch graphs of quadric surfaces.

Definition: In order to assist in sketching the graph of a surface, one can set one of the variables equal to a constant to obtain curves of intersection of the surface with planes parallel to the coordinate planes. These curves are called <u>traces</u> of the surface.

Note: In effect, the above process of finding traces is similar to observing the surface by viewing it while "staring down" a particular axis.

<u>Example 1:</u> Use traces to sketch the quadric surface given by $x^2+\frac{y^2}{4}+\frac{z^2}{9}=1$.

If we stare down the *x*-axis (set $x=\#$), we see $\frac{y^2}{4}+\frac{z^2}{9}=1-\#^2$, which is an ellipse, provided that $-1<x<1$.

Note that if $x=\pm1$ then we have $\frac{y^2}{4}+\frac{z^2}{9}=0$ which can only be true if $y=0$ and $z=0$. Therefore, when $x=\pm1$ the ellipse has shrunk to a point.

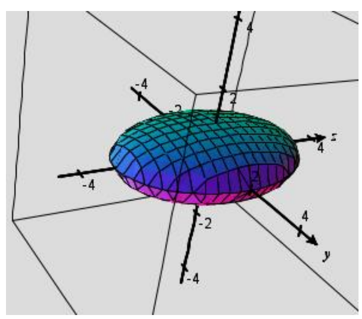
If we stare down the *y*-axis (set $y=\#$), we see $x^2+\frac{z^2}{9}=1-\frac{\#^2}{4}$, which is an ellipse, provided that $-2<y<2$.

Note that if $y=\pm2$ then we have $x^2+\frac{z^2}{9}=0$ which can only be true if $x=0$ and $z=0$. Therefore, when $y=\pm2$ the ellipse has shrunk to a point.

If we stare down the *z*-axis (set $z=\#$), we see $x^2+\frac{y^2}{4}=1-\frac{\#^2}{9}$, which is an ellipse, provided that $-3<z<3$.

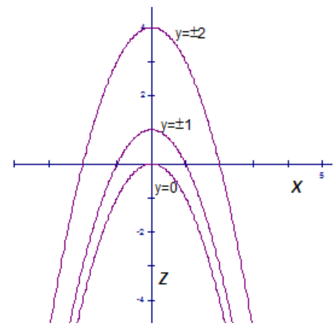
Note that if $z=\pm3$ then we have $x^2+\frac{y^2}{4}=0$ which can only be true if $x=0$ and $y=0$. Therefore, when $z=\pm3$ the ellipse has shrunk to a point.

So, a 3 dimensional sketch is:



Terminology: Since all the traces are ellipses the above surface is called an <u>ellipsoid</u>.

Example 2: Use traces to sketch the surface given by $z = y^2 - x^2$.



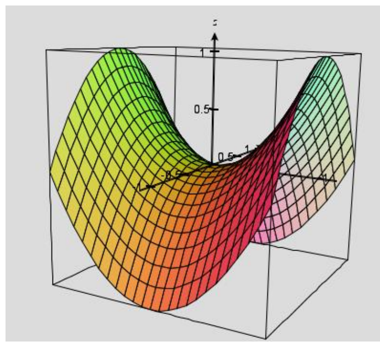
If we stare down the x-axis (set $x = \#$), we see $z = y^2 - \#^2$ which is a parabola.

If we stare down the y-axis (set $y = \#$), we see $z = \#^2 - x^2$ which is a parabola.

If we stare down the z-axis (set $z = \#$), we see $\# = y^2 - x^2$ which is a hyperbola.

Note that if $z < 0$ then we have $x^2 - y^2 = \#$.

A sketch of the surface is:



Terminology: Since the traces of this surface are hyperbolas and parabolas, we call this a <u>hyperbolic paraboloid</u>.

Example 3: Use traces to sketch the surface given by $z = x^2 + y^2 + 1$. Make a guess as to what you think this surface is called.

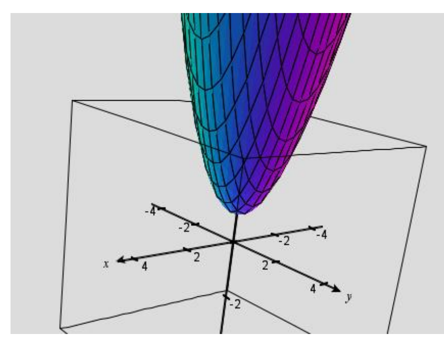
If we stare down the x-axis (set $x = \#$), we see $z = \#^2 + y^2 + 1$ which is a parabola.

If we stare down the y-axis (set $y = \#$), we see $z = x^2 + \#^2 + 1$ which is a parabola.

If we stare down the z-axis (set $z = \#$), we see $x^2 + y^2 = 1 - \#$ which is an ellipse.

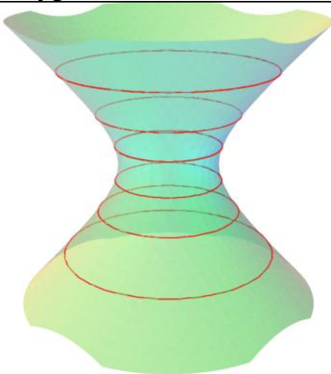
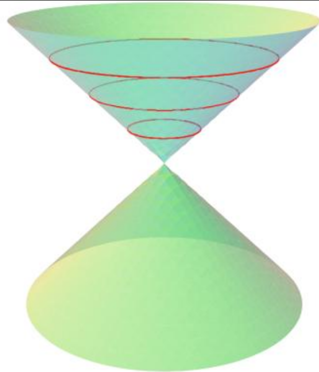
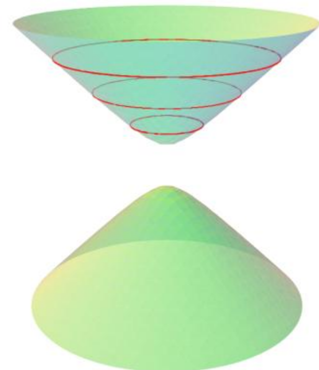
Note that if $z = 1$ we have $x^2 + y^2 = 0$ and therefore the ellipses shrink to a point.

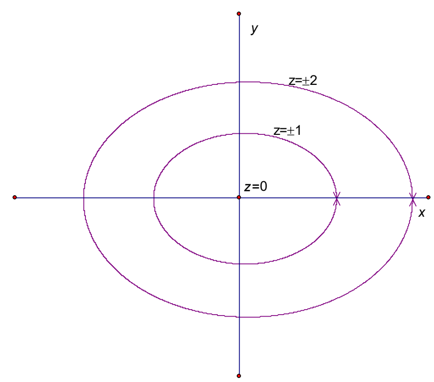
If $z > 1$ then $x^2 + y^2 = negative$ implies that no ellipses exists when $z > 1$.



Terminology: Since the trace curves of this surface are parabolas and ellipses, this quadric surface is called an <u>elliptic paraboloid</u>.

The next three surfaces (cones, hyperboloid of one sheet, and hyperboloid of two sheets) are the most difficult to distinguish because they all have traces of *ellipses* and *hyperbolas*. The key to distinguishing these surfaces from one another is to watch carefully what happens with the *ellipses*.

Hyperboloid of 1 Sheet	**Cone**	**Hyperboloid of 2 sheets**
		
Note that here the ellipses always exist! This is indicated in the equation when our ellipse equation can never be equal to 0 or a negative $\left(x^2+y^2=4\right)$	Note that here the ellipses shrink down to a single point. This is indicated in the equation when our ellipse shrinks having a single solution $\left(x^2+y^2=0\right)$	Note that here the ellipses shrink down to a single point and then disappear for a time! This is indicated in the equation when our ellipse shrinks having a single solution $\left(x^2+y^2=0\right)$ and then fails to have any solutions $\left(x^2+y^2=-1\right)$



<u>Example 4:</u> Use traces to sketch the surface given by $z^2=\frac{x^2}{4}+y^2$.

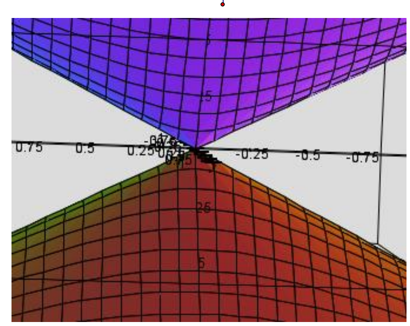
If we stare down the x-axis (set $x=\#$), we see $z^2-y^2=\frac{\#^2}{4}$ which is a hyperbola.

If we stare down the y-axis (set $y=\#$), we see $z^2-\frac{x^2}{4}=\#^2$ which is a hyperbola.

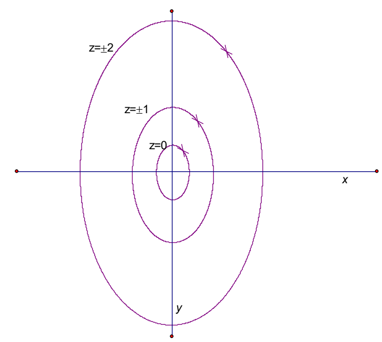
If we stare down the z-axis (set $z=\#$), we see $\#^2=\frac{x^2}{4}+y^2$ which is an ellipse.

Note that if $z=0$ we have $\frac{x^2}{4}+y^2=0$ and therefore the ellipses shrink to a point.

A sketch of the surface is shown at the right.



Terminology: This type of quadric surface is called a <u>cone</u>. Its trace curves are ellipses and hyperbolas (in some cases the hyperbolas are degenerate).



Example 5: Uses traces to sketch the surface given by $x^2 + \frac{y^2}{4} - z^2 = 1$.

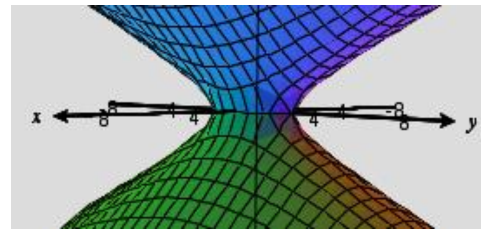
If we stare down the *x*-axis (set $x = \#$), we see $\frac{y^2}{4} - z^2 = 1 - \#^2$ which is a hyperbola.

If we stare down the *y*-axis (set $y = \#$), we see $x^2 - z^2 = 1 - \frac{\#^2}{4}$ which is a hyperbola.

If we stare down the *z*-axis (set $z = \#$), we see $x^2 + \frac{y^2}{4} = 1 + \#^2$ which is an ellipse.

Note that no matter the value of *z* these ellipses will never shrink to a point and will always exist.

A sketch of the surface is shown at the right.



Terminology: This quadric surface is called a <u>hyperboloid of one sheet</u>. Its trace curves are ellipses and hyperbolas.

Example 6: Use traces to sketch the surface given by $-x^2 - \frac{y^2}{4} + z^2 = 4$. Make a guess as to what you think the name of the surface is.

If we stare down the *x*-axis (set $x = \#$), we see $z^2 - \frac{y^2}{4} = 4 + \#^2$ which is a hyperbola.

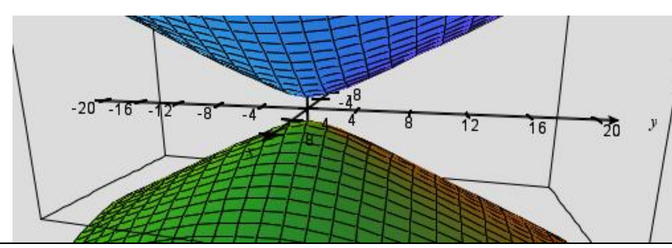
If we stare down the *y*-axis (set $y = \#$), we see $z^2 - x^2 = 4 + \frac{\#^2}{4}$ which is a hyperbola.

If we stare down the *z*-axis (set $z = \#$), we see $x^2 + \frac{y^2}{4} = \#^2 - 4$ which is an ellipse.

Note that if $z = \pm 2$ then $x^2 + \frac{y^2}{4} = 0$ and therefore the ellipses shrink to a point.

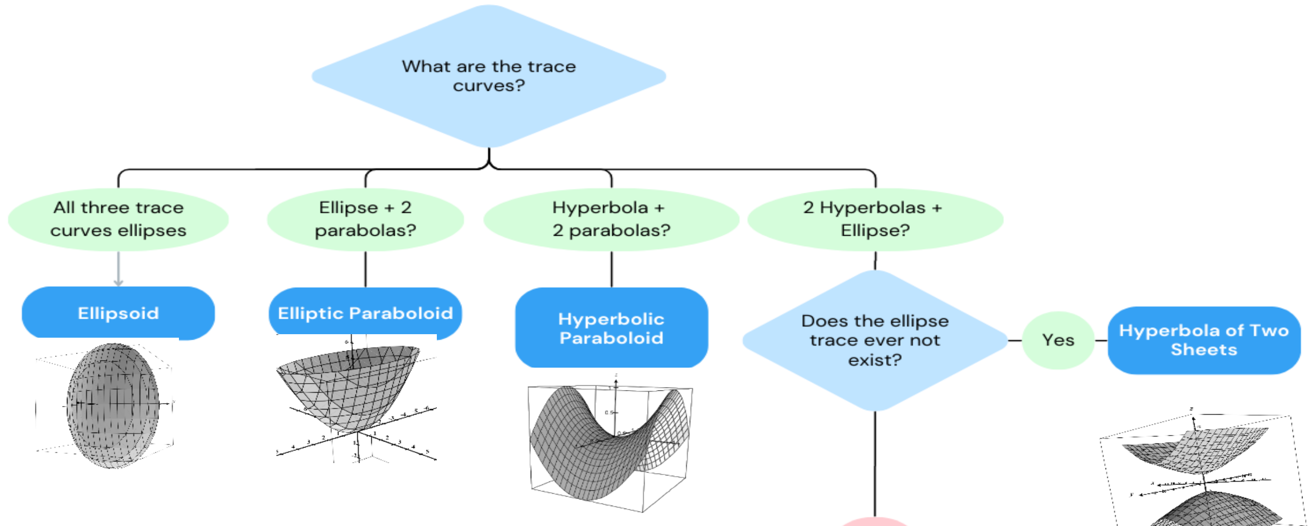
If $-2 < z < 2$ then $x^2 + \frac{y^2}{4} = \textit{negative}$ and therefore the ellipses don't exist.

A sketch of the surface is shown at the right.



Terminology: This quadric surface in the above example is called a <u>hyperboloid of two sheets</u>. Like the hyperboloid of one sheet, its trace curves are ellipses and hyperbolas.

Quadric Classification Decision Tree:



Recall the following conic structures:

- Two squared terms of same sign added together represents an ellipse.
  - Ex. $\frac{x^2}{m}+\frac{y^2}{n}=\#$
- Two squared terms of different signs added together represents a hyperbola.
  - Ex. $\frac{x^2}{m}-\frac{y^2}{n}=\#$
- A squared and linear term of either sign added together represents a parabola.
  - Ex. $\frac{x^2}{m}-\frac{y}{n}=\#$

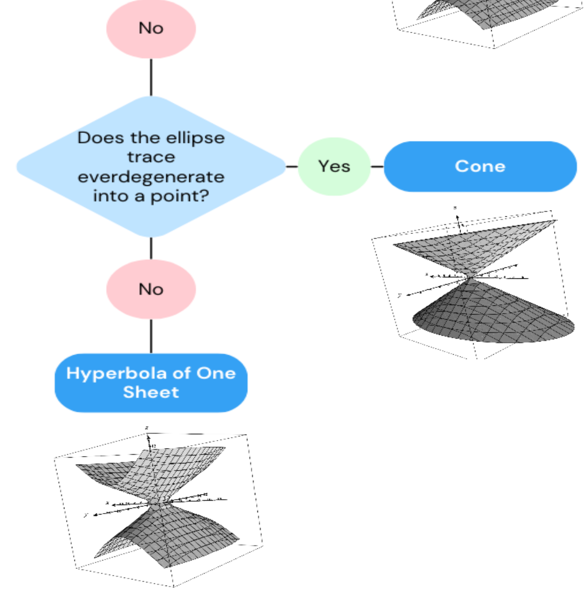


Table of Quadric Surfaces:

Quadric Surface	Trace Curve(s)	Possible Equation Form
Ellipsoid	Ellipse	$\frac{x^2}{a^2}+\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$
Elliptic Paraboloid	Ellipse, Parabola	$\frac{z}{c}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$
Hyperbolic Paraboloid	Hyperbola, Parabola	$\frac{z}{c}=\frac{x^2}{a^2}-\frac{y^2}{b^2}$
Cone	Ellipse, Hyperbola	$\frac{z^2}{c^2}=\frac{x^2}{a^2}+\frac{y^2}{b^2}$
Hyperboloid of One Sheet	Ellipse, Hyperbola	$\frac{x^2}{a^2}+\frac{y^2}{b^2}-\frac{z^2}{c^2}=1$
Hyperboloid of Two Sheets	Ellipse Hyberpola	$-\frac{x^2}{a^2}-\frac{y^2}{b^2}+\frac{z^2}{c^2}=1$

Note: The reason that the table says "Possible Equation Form" is that the roles of x, y, and z may be interchanged. Also, it is possible that shifts along the x, y, or z axis occur. The next example illustrates these ideas.

Example 7: Identify the quadric surface determined by each of the following equations.

a) $-x^2+\frac{y^2}{4}-\frac{z^2}{2}=1$

If we look down the x-axis we see hyperbolas.

If we look down the y-axis we see ellipses $x^2+\frac{z^2}{2}=\frac{y^2}{4}-1$.

These ellipses shrink to a point when $y=\pm 2$ and don't exists on the interval $(-2,2)$.

If we look down the z-axis we see hyperbolas.

Therefore, this equation determines a hyperboloid of two sheets.

b) $(y-1)=\frac{(x-3)^2}{1}+\frac{z^2}{2}$

If we look down the x-axis we see parabolas.

If we look down the y-axis we see ellipses $(y-1)=\frac{(x-3)^2}{1}+\frac{z^2}{2}$.

These ellipses shrink to a point when $y=0$ and don't exist on the interval $(-\infty,1)$.

If we look down the z-axis we see parabolas.

Therefore, this equation determines an elliptic paraboloid.